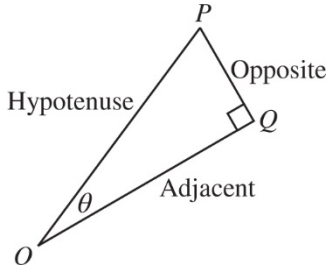
Evaluate Trigonometric Functions of Any Angle

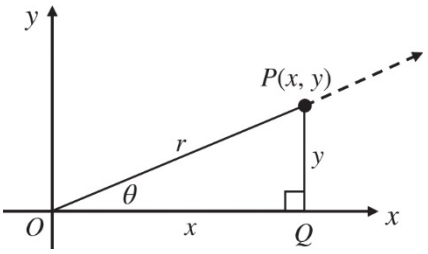
ΔPOQ is a right triangle with acute angle θ.



We can orient the triangle with angle θ in standard position, and label a point $P(x, y)$ on the terminal side of θ.

The length of the horizontal leg is x and the length of the vertical leg is y.

The hypotenuse $r = \sqrt{x^2 + y^2}$ is the distance between $P(x, y)$ and $(0, 0)$.

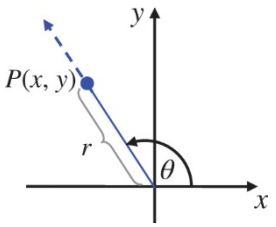


Relative to angle θ, side y is the opposite side, x is the adjacent side, and r is the hypotenuse.

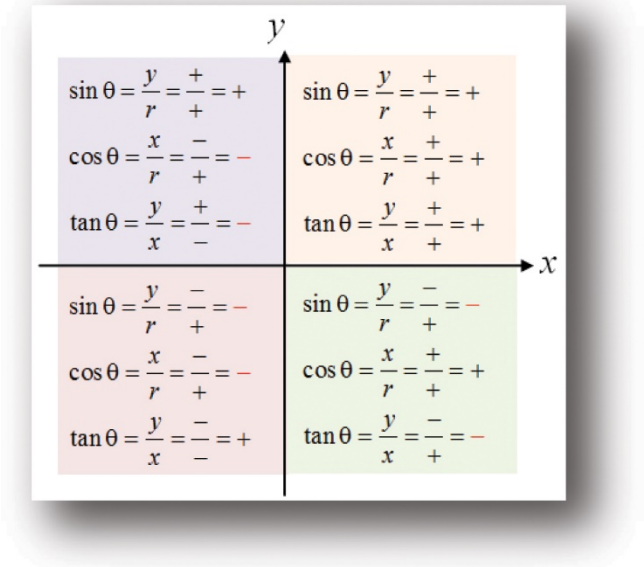
From the right triangle definition of the six trigonometric functions, we have:

$$\sin\theta = \frac{\text{opp}}{\text{hyp}} = \frac{y}{r} \qquad \cos\theta = \frac{\text{adj}}{\text{hyp}} = \frac{x}{r} \qquad \tan\theta = \frac{\text{opp}}{\text{adj}} = \frac{y}{x}$$

$$\csc\theta = \frac{\text{hyp}}{\text{opp}} = \frac{r}{y} \qquad \sec\theta = \frac{\text{hyp}}{\text{adj}} = \frac{r}{x} \qquad \cot\theta = \frac{\text{adj}}{\text{opp}} = \frac{x}{y}$$



These definitions may be extended to angles that are not acute. (Assume none of the denominators are zero.)



The signs of the values of the trigonometric functions depend solely on the signs of x and y.

The reciprocal functions cosecant, secant, and cotangent, respectively will have the same signs as sine, cosine, and tangent.

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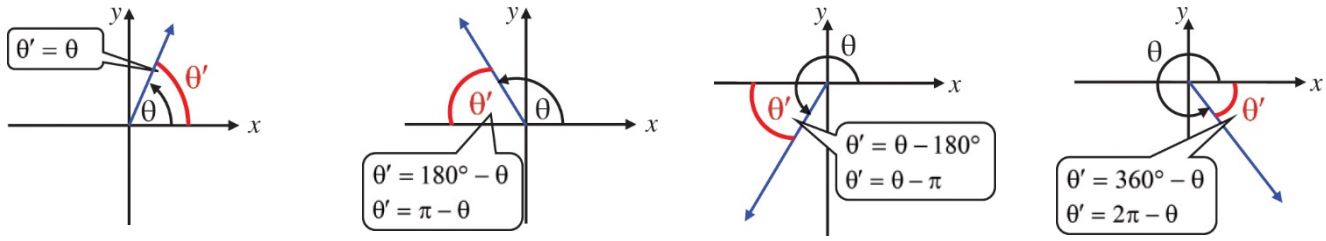
1. Let $P(20,-21)$ be a point on the terminal side of angle θ drawn in standard position. Find the value of $\sin\theta$ and $\cot\theta$.

Angles with terminal sides on the coordinate axes are termed quadrantal angles. In the case of these angles, ratios with $x = 0$ or $y = 0$ in the denominator are undefined.

2. Find the sine, cosine, and tangent for the angle $\theta = 90°$.

Determine Reference Angles

Reference Angle: Let θ be an angle in standard position. The **reference angle** for θ is the acute angle θ' formed by the terminal side of θ and the horizontal axis.



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3. Find the reference angle for the given angle.

a. $225°$ b. $\dfrac{11\pi}{6}$ c. $-\dfrac{2\pi}{3}$ d. $390°$

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Evaluate Trigonometric Functions Using Reference Angles

To find the value of a trigonometric function of a given angle θ,

1. Determine the function value of the reference angle θ'.
2. Affix the appropriate sign based on the quadrant in which θ lies.

Use reference angles to find the exact value.

4. cos 135°

5. sin 240°

6. $\tan\frac{5\pi}{6}$

7. $\csc\left(-\frac{11\pi}{6}\right)$

8. cot 540°

9. $\sec\frac{5\pi}{3}$

10. Find two angles between 0 and 360° where $\cos\theta = -\frac{\sqrt{3}}{2}$.

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11. Given $\sin\theta = -\frac{\sqrt{55}}{8}$ and $\cos\theta < 0$, find $\cot\theta$ and $\cos\theta$.

Pythagorean Identities	Reciprocal Identities	Quotient Identities
$\sin^2\theta + \cos^2\theta = 1$	$\csc\theta = \frac{1}{\sin\theta}$	$\tan\theta = \frac{\sin\theta}{\cos\theta}$
$\tan^2\theta + 1 = \sec^2\theta$	$\sec\theta = \frac{1}{\cos\theta}$	$\cot\theta = \frac{\cos\theta}{\sin\theta}$
$1 + \cot^2\theta = \csc^2\theta$	$\cot\theta = \frac{1}{\tan\theta}$	

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12. Given $\sec\theta = \frac{41}{40}$ for θ in Quadrant IV, use fundamental identities to find tan θ and sin θ.

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